

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Impacts of agriculture, social/political factors & sustainable agriculture - Set 1

Mark scheme

AQA spec 3.5.1.3-3.5.1.5; Biological resources; textbook pp. 328-337

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive (2) • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls (2) • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted (2)	6
01.2	<i>1 mark for each point.</i> • Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows (1) • These communities depend on the farming continuing (1)	2
01.3	<i>1 mark for each point.</i> • CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration (1) • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation (1) • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase (1) • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared (1)	3
02.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes (2) • Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons (2)	4
02.3	<i>1 mark for each point.</i> • Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock (1) • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output (1) • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
03.1	<i>1 mark for each point.</i> • Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers (1) • Eastern Europe / USSR needed food but could not pay (1) • Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production (1)	3
03.2	<i>1 mark for each point.</i> • Quotas limiting production (eg milk) (1) • Set-aside payments for taking land out of production (1) • Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) (1) • Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming (1)	4
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape (1) • Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat (1) • Beetle banks of tussocky grass - natural pest predators (1) • Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators (1) • In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
	Model answer - Level 3 (8 marks)	
	Sustainability matters because the population may reach 12 billion, land has been degraded since farming began in the fertile crescent, and current yields depend on artificial fertilisers, pesticides and irrigation that draws down groundwater, with little new land to cultivate. Pest control can move from chemicals to cultural methods - weeding, mulching, rotation, barrier and companion crops, habitats for predators - biological control, sterile males and pheromones, resistant varieties and integrated control, with fewer antibiotics in livestock.	

Question	Marking guidance	Marks
	Nutrients can come from natural processes: legumes fixing nitrogen, recycling manure and compost, rotation, permaculture and conserving soil biota and mycorrhizae, reducing manufactured nitrate and the non-renewable rock phosphate it accompanies. Energy use falls with low tillage, less machinery, renewable power, low food miles and lower methane from cattle diets and drought-tolerant rice. Water is conserved with low water-use crops, drip irrigation, reservoirs and aquifer recharge and by avoiding saline water; soil by maintaining organic matter and controlling erosion. Retaining hedgerows, ditches, ponds and woodland, conserving crop wild relatives in seed banks and agri-environment schemes protect the biodiversity that supplies pollination and pest control, while Fairtrade, local food and consumer choices address the social side.	
	The tension is between yield and sustainability: some strategies cost more or yield less in the short term and need policy support to be adopted. But since the unsustainable inputs themselves depend on finite fossil fuels and phosphate, the strategies that use natural processes are the only ones that can feed a larger population indefinitely.	
	Weaker answer - Level 2 (4 marks)	
	Agriculture can be made more sustainable by using less pesticides and using biological control instead, using manure instead of artificial fertilisers, growing legumes which add nitrogen to the soil, and using crop rotation. Drip irrigation saves water. Farmers can keep hedgerows to provide habitats for wildlife. Organic farming uses these methods. These strategies reduce pollution and keep the soil healthy so food can be produced in the future.	
	What makes the difference	
	• Level 2 lists valid strategies but does not explain why they are needed (population growth, land degradation, dependence on finite inputs) or group them by the problem they solve.	
	• Level 3 covers energy, soil, biodiversity and social strategies as well as pests, nutrients and water.	
	• Level 3 discusses the yield-sustainability tension and the need for policy support; Level 2 has no evaluation.	
04.2	<i>1 mark.</i> • Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
05.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hedgerows (1) • Hay meadows (1) • Grazed moorland / heathland (1) • Chalk grassland (1)	2
05.2	<i>1 mark for each point.</i> • Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive (1) • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers (1)	2

Question	Marking guidance	Marks
05.3	<i>1 mark.</i> The introduced (non-indigenous) predator may eat other species as well as the pest (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
06.1	<i>1 mark for each point.</i> - Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed (1) - Yield mapping of different parts of a field controls fertiliser application in each area for future crops (1) - This maximises yield while avoiding wastage / reducing leaching and nutrient pollution (1)	3
06.2	<i>1 mark for each point.</i> - Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive (1) - They are mainly available in affluent societies; poorer farmers cannot afford the inputs (1)	2
06.3	<i>1 mark.</i> Provides an income that allows producers to afford basic rights such as water, education, health care and food (1)	1
06.4	<i>1 mark for each point.</i> - Nitrates are leached into water bodies used as sources of potable water (1) - High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic (1)	2
06.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Anaerobic digestion by microbes in rice padi fields (1) - Livestock intestines (ruminant digestion) (1) - Anaerobic decay of manure / slurry (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
07.1	<i>1 mark for each point.</i> - About 10% of land is already cultivated and 25% grazed (1) - Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil (1)	2
07.2	<i>1 mark for each point.</i> - Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies (1) - So people with less money lose access to the land and food they used to produce (1)	2
07.3	<i>1 mark for each point.</i> - It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services (1) - So that pesticides and artificial fertilisers become unnecessary rather than just being banned (1)	2

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07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	